

# Assignment no # 1

Section – A

Software Engineering

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# Library Management System (LMS)

Version 1.0

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| Field          | Details                                                         |
|----------------|-----------------------------------------------------------------|
| Document Title | Software Requirements Specification – Library Management System |
| Version        | 1.0                                                             |
| Status         | Draft                                                           |
| Prepared By    | Group 7                                                         |
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## 1. Introduction

### 1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Library Management System (LMS). The document serves as the authoritative reference for developers, testers, project managers, and stakeholders throughout the software development lifecycle.

### 1.2 Scope

The Library Management System is a web-based application designed to automate and streamline library operations. It covers:

- Cataloging and managing all library resources (books, journals, digital media)
- Member registration, management, and access control
- Book issuing, returning, and renewal workflows
- Fine calculation and payment tracking
- Search and discovery of library resources
- Reporting and analytics for library administrators

1.3 Definitions & Acronyms

| Term          | Definition                                                                                |
|---------------|-------------------------------------------------------------------------------------------|
| LMS           | Library Management System                                                                 |
| Member        | Registered library user with borrowing privileges                                         |
| Librarian     | Staff member managing day-to-day operations including circulation and resource cataloging |
| Administrator | System admin with full access and configuration rights                                    |
| ISBN          | International Standard Book Number                                                        |
| OPAC          | Online Public Access Catalog — public-facing search interface                             |
| SRS           | Software Requirements Specification                                                       |
| UI            | User Interface                                                                            |
| API           | Application Programming Interface                                                         |

1.4 Intended Audience

- Software developers and architects building the system
- QA engineers designing test plans
- Project managers overseeing delivery
- Library staff (Librarians and Administrators) who will use the system
- Stakeholders reviewing project scope and approval

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2. Overall Description

2.1 Product Perspective

The LMS is a standalone web application that can optionally integrate with external systems such as payment gateways, barcode/RFID scanners, and email/SMS notification services. It replaces manual, paper-based library processes with a centralized digital platform accessible via any modern browser.

2.2 User Classes & Characteristics

| User Type       | Description                                                         | Technical Level |
|-----------------|---------------------------------------------------------------------|-----------------|
| Member / Patron | Students, faculty, or public users who search and borrow books      | Low             |
| Librarian       | Staff who issue, receive, return, and catalog all library resources | Medium          |

| User Type     | Description                                                   | Technical Level |
|---------------|---------------------------------------------------------------|-----------------|
| Administrator | IT/library managers who configure the system and manage users | High            |

2.3 Assumptions & Dependencies

- The system will be deployed on a web server accessible via HTTPS.
- Users will access the system using modern browsers (Chrome, Firefox, Safari, Edge — latest 2 versions).
- A stable internet connection is required for web-based access.
- The library will provide barcode scanners or RFID readers for physical book check-in/out (optional hardware).
- Email/SMS gateway credentials will be provided for notification services.
- Initial data migration from existing records (if any) is a separate workstream.

2.4 Constraints

- The system must comply with applicable data privacy regulations (e.g., GDPR, local data protection laws).
- The system must support a minimum of 10,000 book records and 5,000 active members.
- Response time for search queries must not exceed 2 seconds under normal load.
- All financial transactions must be logged and auditable.

3. Functional Requirements

3.1 User Authentication & Authorization

| ID         | Requirement                                                                                                                        | Priority |
|------------|------------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-AUTH-01 | The system shall support role-based login for Members, Librarians and Administrators.                                              | High     |
| FR-AUTH-02 | Users shall authenticate using email/username and password. Optional two-factor authentication (2FA) must be supported on sign up. | High     |
| FR-AUTH-03 | The system shall enforce session timeout after 15 minutes of inactivity.                                                           | Medium   |
| FR-AUTH-04 | Administrators shall be able to create, edit, deactivate, and delete user accounts.                                                | High     |
| FR-AUTH-05 | Password reset via registered email or phone number shall be supported.                                                            | High     |
| FR-AUTH-06 | The system shall log all login attempts (successful and failed) with timestamps.                                                   | Medium   |

### 3.2 Book & Resource Catalog Management

| ID        | Requirement                                                                                                                                                   | Priority |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-CAT-01 | Librarians shall be able to add new books/resources with details: title, author(s), ISBN, publisher, edition, year, category, language, and number of copies. | High     |
| FR-CAT-02 | The system shall support import of book data via CSV or via ISBN lookup (external API integration).                                                           | Medium   |
| FR-CAT-03 | Librarians shall be able to update, archive, and delete resource records.                                                                                     | High     |
| FR-CAT-04 | The system shall track the total copies, available copies, and copies currently issued for each resource.                                                     | High     |
| FR-CAT-05 | The system shall support multiple resource types: books, journals, magazines, DVDs, and e-resources.                                                          | Medium   |
| FR-CAT-06 | Each resource record shall have a unique barcode/accession number auto-generated.                                                                             | High     |

### 3.3 Member Management

| ID        | Requirement                                                                                                                 | Priority |
|-----------|-----------------------------------------------------------------------------------------------------------------------------|----------|
| FR-MEM-01 | Librarians shall be able to register new members with: name, ID/card number, contact info, address, member type, and photo. | High     |
| FR-MEM-02 | The system shall generate a unique membership ID and library card number for each member.                                   | High     |
| FR-MEM-03 | Member accounts shall have an expiry date. The system shall notify members 30 days before expiry.                           | Medium   |
| FR-MEM-04 | Librarians shall be able to renew, suspend, or terminate membership.                                                        | High     |
| FR-MEM-05 | The system shall maintain a complete borrowing history per member.                                                          | High     |
| FR-MEM-06 | Members shall be able to view their own profile, borrowing history, and outstanding fines via the member portal.            | Medium   |

### 3.4 Book Issuing & Returns

| ID        | Requirement                                                                                                 | Priority |
|-----------|-------------------------------------------------------------------------------------------------------------|----------|
| FR-ISS-01 | Librarians shall issue books to members by scanning the barcode or searching by title/member ID.            | High     |
| FR-ISS-02 | The system shall enforce borrowing limits based on member type (e.g., students: 2 books, faculty: 4 books). | High     |
| FR-ISS-03 | The system shall set and display the due date upon issuing a                                                | High     |

| ID        | Requirement                                                                                                  | Priority |
|-----------|--------------------------------------------------------------------------------------------------------------|----------|
|           | book.                                                                                                        |          |
| FR-ISS-04 | Members shall be able to request renewals via the member portal (subject to librarian approval).             | Medium   |
| FR-ISS-05 | The system shall process book returns, update availability in real time, and calculate any applicable fines. | High     |
| FR-ISS-06 | The system shall prevent issuing books to members with overdue items or outstanding fines.                   | High     |
| FR-ISS-07 | The system shall support book reservations/holds for items currently checked out.                            | Medium   |

### 3.6 Search & OPAC

| ID        | Requirement                                                                                                                      | Priority |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-SCH-01 | All users (including unauthenticated visitors) shall be able to search the catalog by title, author, ISBN, subject, and keyword. | High     |
| FR-SCH-02 | Search results shall display availability status (Available / Checked Out / Reserved).                                           | High     |
| FR-SCH-03 | Advanced filters shall include: category, language, publication year range, and resource type.                                   | Medium   |
| FR-SCH-04 | The system shall support autocomplete suggestions in the search bar.                                                             | Low      |
| FR-SCH-05 | Members shall be able to place a hold on a book directly from search results.                                                    | Medium   |

### 3.8 Reporting & Analytics

| ID        | Requirement                                                                                                     | Priority |
|-----------|-----------------------------------------------------------------------------------------------------------------|----------|
| FR-REP-01 | Librarian shall access pre-built reports: Most Borrowed Books, Overdue Items, Active Members, Fine Collections. | High     |
| FR-REP-02 | Reports shall be exportable in PDF and Excel formats.                                                           | High     |

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## 4. Non-Functional Requirements

### 4.1 Performance

- Search queries shall return results within 2 seconds for databases up to 50,000 records under normal load.
- The system shall support at least 200 concurrent users without performance degradation.

- Page load time shall not exceed 3 seconds on a standard broadband connection.
- Batch report generation shall complete within 30 seconds.

## 4.2 Security

- All data in transit shall be encrypted using TLS 1.2 or higher (HTTPS only).
- Passwords shall be stored using a strong hashing algorithm (bcrypt or Argon2).
- The system shall implement protection against SQL injection, XSS, CSRF, and other OWASP Top 10 vulnerabilities.
- Access to administrative functions shall be restricted by IP allowlist (optional, configurable).
- Sensitive data (member PII) shall be encrypted at rest.
- All administrative actions shall be logged in an immutable audit trail.

## 4.3 Reliability & Availability

- The system shall achieve a minimum uptime of 99.5% (excluding scheduled maintenance).
- Scheduled maintenance windows shall be communicated at least 24 hours in advance.
- Automated database backups shall run daily, with backups retained for 30 days.
- The system shall recover from a server crash and be back online within 15 minutes (RTO).

## 4.4 Usability

- The interface shall be intuitive enough for a librarian with basic computer literacy to complete common tasks within 2 hours of training.
- The system shall be fully responsive and usable on desktops, tablets, and mobile browsers.
- The system shall comply with WCAG 2.1 Level AA accessibility standards.
- All error messages shall be descriptive and guide the user toward resolution.

## 4.5 Scalability

- The system architecture shall support horizontal scaling to handle growth in users and data volume.
- The database shall support up to 500,000 resource records and 100,000 member records without structural changes.

## 4.6 Maintainability

- The codebase shall follow documented coding standards and include inline documentation.
- The system shall use a modular architecture to allow independent updates to modules (catalog, members, fines, etc.).
- A deployment pipeline (CI/CD) shall be set up for automated testing and releases.

## 5. System Interfaces

### 5.1 External Integrations

| Interface                                       | Type     | Purpose                                    | Priority |
|-------------------------------------------------|----------|--------------------------------------------|----------|
| Email Gateway (SMTP / SendGrid)                 | API      | Send automated notifications to members    | High     |
| SMS Gateway (Twilio / similar)                  | API      | Send SMS alerts for overdue and holds      | Medium   |
| ISBN Lookup API (Google Books / Open Library)   | REST API | Auto-fill book details on catalog entry    | Medium   |
| Payment Gateway (Mobile Account / Bank Account) | API      | Process digital fine payments              | Low      |
| Barcode / RFID Scanner                          | Hardware | Quick scan for issuing and returning books | Medium   |

### 5.2 Technology Stack (Recommended)

| Layer          | Recommended Technology                                     |
|----------------|------------------------------------------------------------|
| Frontend       | React.js (responsive web UI)                               |
| Backend        | Node.js (Express) or Python (Django/FastAPI)               |
| Database       | PostgreSQL (primary), Redis (caching & session management) |
| Authentication | JWT tokens + OAuth 2.0                                     |
| Hosting        | Cloud (AWS) or on-premises server                          |
| Search         | Elasticsearch or PostgreSQL Full-Text Search               |

## 6. Key Use Cases Summary

| Use Case ID | Name           | Actor                      | Brief Description                                                             |
|-------------|----------------|----------------------------|-------------------------------------------------------------------------------|
| UC-01       | Member Login   | Member / Librarian / Admin | User logs in with credentials; system validates and grants role-based access. |
| UC-02       | Search Catalog | Any User                   | User enters search terms; system displays matching results with               |



| Use Case ID | Name                | Actor              | Brief Description                                                                          |
|-------------|---------------------|--------------------|--------------------------------------------------------------------------------------------|
|             |                     |                    | availability.                                                                              |
| UC-03       | Issue Book          | Librarian          | Librarian scans book and member card; system issues book and sets due date.                |
| UC-04       | Return Book         | Librarian          | Librarian scans returned book; system updates availability and calculates fine if overdue. |
| UC-05       | Reserve Book        | Member             | Member places a hold on a checked-out book; system notifies when available.                |
| UC-06       | Pay Fine            | Member / Librarian | Member or librarian records fine payment; system marks fine as settled.                    |
| UC-07       | Add Book to Catalog | Librarian          | Librarian enters or imports book details; system creates and stores catalog record.        |
| UC-08       | Generate Report     | Administrator      | Admin selects report type and date range; system generates and exports report.             |

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## 7. Acceptance Criteria

The system shall be considered ready for release when all of the following are met:

- All High-priority functional requirements pass their corresponding test cases.
  - System handles 200 concurrent users with search response time under 2 seconds (verified by load testing).
  - Security audit confirms no critical or high-severity OWASP Top 10 vulnerabilities.
  - All automated notifications are delivered within 5 minutes of the triggering event.
  - User Acceptance Testing (UAT) is completed by at least 3 librarians with a satisfaction score of 4/5 or higher.
  - All reports export correctly in PDF and Excel formats.
  - Data backup and restore procedure is tested and documented.
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## 8. Appendix

### 8.1 Requirement Priority Legend

| Priority | Definition                                                                                                 |
|----------|------------------------------------------------------------------------------------------------------------|
| High     | Core functionality — must be included in the initial release. System is unusable without it.               |
| Medium   | Important — should be included in the initial release. Workarounds exist but are not acceptable long-term. |
| Low      | Nice to have — can be deferred to a future release without impacting core operations.                      |

— End of Document —